Revision Notes for Class 10 Science

Chapter 7 – How Do Organisms Reproduce?

1. Do Organisms Create Exact Copies of Themselves?

DNA and Reproduction

- Similar Body Designs: Organisms with similar appearances suggest shared design sources, indicating that reproduction creates these designs.
- Chromosomes and DNA: The nucleus contains chromosomes made of DNA, which carries hereditary information and instructions for protein synthesis.
- DNA Copying: Reproduction involves creating a copy of DNA, which is crucial for passing traits to the next generation. This copying process also involves replicating cellular machinery.
- Cell Division: DNA replication is followed by cell division, resulting in two cells with their own set of DNA and cellular apparatus.
- Variations and Reliability: DNA copying is a biochemical process that can introduce variations. While cells may not be identical to the original, they remain similar. Non-viable DNA copies can lead to cell death, ensuring that only cells with functional DNA survive.

1.1 The Importance of Variation

- Maintaining Body Design: Consistent DNA copying during reproduction ensures that organisms retain their specific body designs and effectively occupy their ecological niches.
- Stability and Adaptation: Reproduction plays a crucial role in maintaining population stability. Variations among individuals become important when environmental changes occur.
- Survival Through Variation: If an environment changes drastically, organisms with variations may have a better chance of survival. For example, if bacteria living in temperate waters face rising temperatures due to global warming, some heat-resistant

variants might survive and thrive, while others may not. Without such variations, the entire species could face extinction.

- Importance of Variation: Variations allow species to adapt to new conditions and continue to exist over time, highlighting the role of genetic diversity in survival and adaptation.

2. Modes of Reproduction

Reproduction can be defined as a process that involves the production of offspring by a particular individual or individuals to propagate their species. Generally, reproduction happens during the reproductive phase of an organism. The mode of reproduction may vary in organisms. They can be broadly categorised as:

Asexual Mode of Reproduction:

- The mode of reproduction employing which a single individual creates a new generation of species is termed asexual reproduction.
- Generally, unicellular organisms exhibit an asexual mode of reproduction, though some of them exhibit a sexual mode too.

Sexual Mode of Reproduction:

- The mode of reproduction using which two individuals take part in the creation of a new generation of species is termed sexual reproduction.

Types of Asexual Mode of Reproduction:

2.1.Fission:

- In unicellular organisms, new individuals are created by the process of cell division or fission.
- The nucleus of the cell divides into new individual cells under favourable conditions.
- Fission can be of two types depending on the number of new individuals created.

a. Binary Fission: This division leads to the formation of two new individuals. These can be further divided based on their plane of division:

i. Irregular binary fission: In this type of fission the plane of division of a cell is irregular, it can be in any plane. Example - Amoeba.

ii. Transverse binary fission: In this type of fission the cells divide along a transverse plane. Example - Paramecium.

iii. Longitudinal binary fission: In this type of fission the plane of cell division is longitudinal. Example - Euglena.

b. Multiple fission: This is the Division of a Single Cell into Many New Daughter Cells. Example - Plasmodium.

2.2.Fragmentation:

- This is a process where an organism simply breaks up into smaller pieces when they are mature.
- Each of the fragments or broken pieces grows into a new individual. There should be a cell that is capable of growing into a new individual in such organisms.
- Example - Spirogyra.

2.3. Regeneration:

- This is a process where some fully differentiated organisms can be cut or broken into pieces and each of their body parts can grow into a new individual.
- Different cells in this mass of cut cells undergo a lot of changes in an organised manner to become different cells and tissues.
- Example - Planaria, Hydra.

2.4. Budding:

- This is a process where a protuberance-like outgrowth which is called a bud grows by repeated cell division at a specific site and then detaches from the parent body to develop into a separate individual organism.
- Example - Hydra.

2.5.Vegetative Propagation

This is the mode of reproduction by which plants reproduce asexually. In this mode, new plants are developed from a plant's vegetative parts like stem, leaf, and root. Different methods of vegetative propagation are carried out in plants which are as follows:

a. Stem Cutting: This involves cutting the stem into small pieces having internodes and axillary buds. These are then planted in the soil to propagate into new plants. This method is used in sugarcane, hibiscus, drumsticks, etc.

b. Layering: This is a method where the young stem of a plant is bent and buried in the soil to develop roots and thus a new plant. Once the new plant develops, the stem is detached from the parent plant. This is used in jasmine and bougainvillaea.

c. Grafting: This is a method wherein the stems of two different plants are cut and joined together to unite and start developing into a new plant. This is used in nutmeg, roses, etc.

d. Leaf buds: This is a method in which the buds in the notches of leaves develop into new plants. This can be seen in bryophyllum.

Advantages of the Vegetative Propagation, are as follows:

a. The plants that are grown by vegetative propagation bear flowers and fruits earlier as compared to the plants produced from seeds.

b. All plants that are produced this way are genetically similar to the parent plant and have all its characteristics.

2.6. Spore Formation:

- Many multicellular organisms have specific reproductive parts.
- They have tiny thread-like structures with a blob called sporangia.
- These contain cells or spores that eventually develop into new individuals. The spores are very light and covered by a thick wall to protect them and when they come in contact with a moist surface they start to grow.
- Example - Rhizopus.

3. Sexual Reproduction:

3.1. Why Sexual Reproduction?

- Involves Two Organisms: Sexual reproduction requires both a male and female to produce offspring, combining their genetic material.
- Genetic Variation: This mode of reproduction creates greater genetic diversity by mixing DNA from two individuals, resulting in unique combinations of traits.
- Gene Pool Mixing: It enhances the genetic variability within a population, contributing to the survival and adaptability of the species.

- DNA Management: Each parent contributes germ cells (gametes) with half the number of chromosomes. When combined, these restore the full chromosome count in the offspring.
- Specialised Gametes: In complex organisms, gametes are specialised—female gametes are large and nutrient-rich, while male gametes are small and motile. This specialisation influences the reproductive systems and body differences between males and females.

3.2. Sexual Reproduction in Flowering Plants

- The process of sexual reproduction in plants involves the fusion of gametes to produce offspring. The reproductive parts in angiosperms (plants that flower and produce fruits and seeds) are located in the flower. The parts of a flower consist of sepals, petals, stamens, and pistils.
- The reproductive parts of the flower are the stamen which contains the male gamete and the pistil containing the female gametes.

  a. Stamen: This is the male reproductive part and is also known as the androecium. It consists of a filament and an anther that encloses the pollen grains. The pollen grains produce the male germ cells or gametes.

  b. Pistil: This is the female reproductive part of the flower and is also known as gynoecium. This is made of three parts, namely, stigma, style, and ovary. The enlarged portion at the bottom of a pistil is the ovary which contains an ovule with an egg cell. The middle long part of the pistil is the style and the terminal sticky part is the stigma.
- **Based on the presence of the stamen or pistil, flowers can be classified as:**

  a. Unisexual: These are the flowers that contain either stamens or pistils. These are also called incomplete flowers. Examples - papaya, mulberry, watermelon, etc.

  b. Bisexual: These are flowers that contain stamens as well as pistils. Examples - Hibiscus, mustard, rose, etc.
- **The process of sexual reproduction in plants starts with the fusion of the male and the female gametes, followed by the formation of a zygote that eventually develops into a new plant. The process is explained as follows:**

  a. Pollination:

- The process of sexual reproduction in plants starts with the transfer of pollen grains from the anther of the stamen to the stigma of the pistil. This process is termed pollination.
- This is facilitated by pollinating agents like wind, birds, animals, water, etc. which transfer the pollen grains.
- **There can be two types of pollination as follows:**

i. Self-pollination: This involves the transfer of pollen grains from the anther to the stigma of the same flower. Example - wheat, peanut, etc.

ii. Cross-Pollination: This type of pollination involves the transfer of the pollen grains from the anther of one flower to the stigma of another flower of the same species. Example - apples, pumpkins, etc.

b. Fertilisation:

- Through the process of pollination, the pollen is deposited in the style of the pistil. For the next process in reproduction, it needs to reach the female germ cells which are present in the ovary.
- To facilitate this, a tube grows out of the pollen grain and reaches the ovule in the ovary of the pistil.
- Here in the ovule the male germ-cell fuses with a female germ-cell to form a zygote. This process of fusion of the gametes is termed fertilisation.
- After the process of fertilisation, the zygote thus forms and divides repeatedly to form an embryo inside the ovule. The ovule later develops into a seed.
- Meanwhile, the ovary grows and ripens into a fruit and the other parts of the flower, namely the petals, sepals, stamens, style and stigma may be shed off.
- The seed present inside the fruit encloses the future plant in its embryo.

c. Germination:

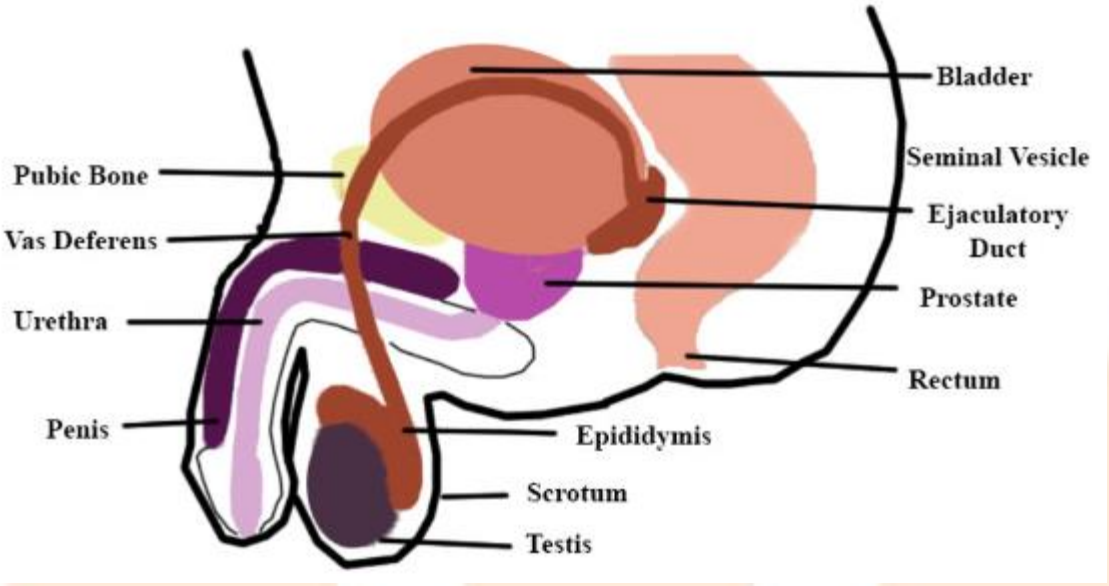
- The seed that contains the new plant or embryo develops into a seedling when the conditions are suitable. This process is termed as germination. Certain conditions like nutrients, water, and proper temperature are necessary for the process of germination.
- The embryo gets its food from the reserve food material stored in the cotyledons. It also has a protective outer covering known as a seed coat.

3.3. Reproduction in Human Beings.

- Sexual Reproduction: Humans reproduce sexually, involving the union of male and female germ cells to create offspring.
- Reproductive Phase: This phase begins when individuals reach sexual maturity, typically marked by adolescence.
- Adolescence and Puberty: During puberty, the body changes to prepare for reproduction. This period includes the development of specialised germ cells necessary for sexual reproduction.
- Changes in Boys: Boys experience facial and body hair growth, voice changes, increased sweat and oil production, and penile enlargement.
- Changes in Girls: Girls see growth of pubic hair, breast development, oily skin with pimples, and the onset of menstruation.
- Specialised Organs: Sexual reproduction requires specific organs, such as the penis in males and the uterus in females, to transfer and nurture germ cells and support pregnancy.

3.3.1. Male Reproductive System

- The male reproductive system consists of organs that produce and transport the male germ cell or gamete, the male hormone testosterone, and the organs that facilitate the discharge of male germ cells into the female reproductive system for fertilisation.
- The male gamete is the sperm which is a tiny body containing the genetic material and they have a long tail for motility to help them reach the female germ-cell for fertilisation.
- The system consists of some external organs like a penis, scrotum, and testes and internal organs like urethra, prostate and seminal vesicles.



a. Testes: The Testes is the part that is responsible for the production of the male germ cell or sperm and the male hormone testosterone. Testes are present in a structure known as the scrotum, located outside the abdominal cavity. This is thus located because the formation of sperm requires a temperature that is lower than the normal body temperature. The hormone testosterone plays a role in regulating the formation of sperm and also the development of secondary sexual characteristics that are seen in boys during puberty.

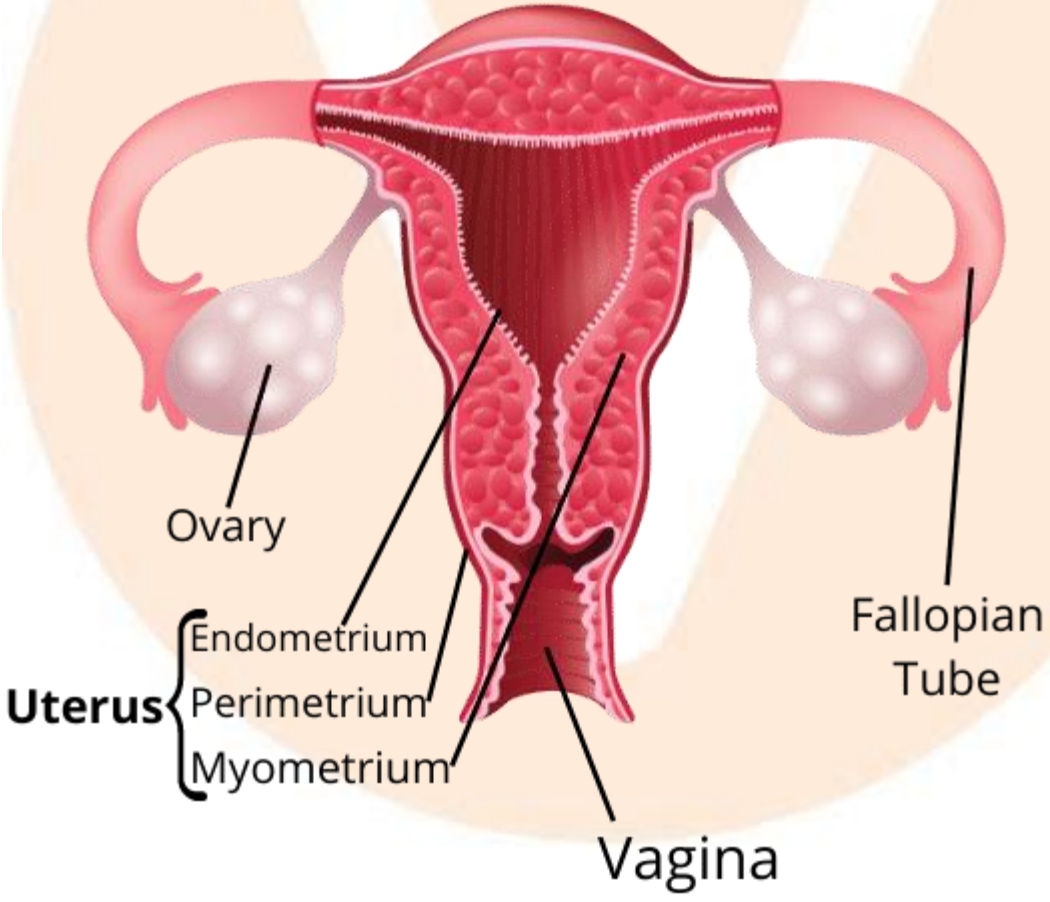
b. Vas Deferens: The sperm that are produced in the testes are stored in the epididymis. Vas deferens is a tube that transports these sperm to the urethra.

c. Urethra: This is a common passage for the sperm as well as urine. The same passage connects the urinary bladder and the vas deferens.

d. Prostate Gland and Seminal Vesicles: These glands are located along the vas deferens. They secrete a fluid, called semen that nourishes the sperm. This semen helps in the easier movement of sperm.

3.3.2. Female Reproductive System.

- The female reproductive system includes the organs that produce the female germ cells and provide a site for fertilisation of the gametes and the development of the embryo into a new individual.
- The female gametes are the eggs that are produced in the ovaries.
- They also produce some hormones like oestrogen and progesterone that are responsible for the onset of secondary sexual characteristics in girls at puberty.
- This system includes a pair of ovaries, a pair of oviducts, a uterus, and a vagina that opens externally through the urethra.



- Ovaries: The ovaries are a pair of glands that are located on either side of the uterus. The ovaries protect the female gametes or eggs and make them suitable for fertilisation. At birth, the ovary of a girl contains thousands of eggs that are immature. After puberty, when the eggs mature, the ovaries release one egg every month. The ovaries also produce the hormones oestrogen and progesterone that are essential in bringing secondary sexual changes in a girl at puberty.
- Fallopian tube: This is also known as the oviduct. This is a thin tube that connects the ovaries to the uterus. The eggs that are released by the ovary are transported through this tube.
- Uterus: This is a bag-like muscular elastic structure into which the two oviducts open. The uterus is the site where the fertilised egg is implanted and grows into a foetus. It is made of 3 tissues, the outer perimetrium, the middle layer of myometrium, and the inner endometrium. This is also responsible for supporting the developing foetus during the entire gestation period.
- Cervix: This is the site where the uterus opens into the vagina. This facilitates a passage for the entry of the sperm into the uterus.

Fertilisation and Development

- Fertilisation: Sperm enters the female reproductive system through the vagina during intercourse and travels through the uterus to the fallopian tubes, where it meets and fertilises the egg.
- Zygote Formation: The fertilised egg, or zygote, starts dividing and moves down the fallopian tube to the uterus.
- Embryo Development: The zygote implants in the uterine lining, where it grows into an embryo.
- Placenta Role: The placenta connects the embryo to the mother, providing nutrients and oxygen while removing waste.
- Gestation Period: The embryo develops into a foetus over about nine months, with birth occurring through the rhythmic contractions of the uterine muscles.

3.3.3. What Happens When the Egg is Not Fertilised?

- An egg is released by the ovary every month in anticipation of it getting fertilised. In case the egg does not get fertilised, it can survive for only a day. Similar to the ovary releasing an egg every month, every month, the uterus too prepares itself for the fertilised egg by creating a thick and spongy lining to provide nourishment to the embryo.
- When fertilisation does not occur, this lining too is not required and this lining and the egg are shed as blood and mucous through the vagina. This is called menstruation. This cycle occurs every month and lasts for about 2 - 8 days roughly.

3.3.4. Reproductive Health.

Sexual Maturation and Reproductive Health

- Sexual Maturation: This gradual process occurs during overall body growth but does not immediately prepare individuals for sexual activity, marriage, or parenthood.
- Reproductive Health: Focuses on safe and healthy sexual practices. It helps young people make informed decisions amidst peer, family, and societal pressures.
- Sexually Transmitted Diseases (STDs): Infections such as gonorrhoea, syphilis, warts, and HIV can result from unsafe sexual practices. These diseases can cause severe health complications if untreated.
- **Contraceptive Methods:**
    - Physical Barriers: Condoms prevent sperm from reaching the egg.
    - Contraceptive Devices: IUCDs like Copper-T are implanted in the uterus to block sperm.
    - Hormonal Methods: Pills like Mala D and I-pill alter the hormonal balance to prevent fertilisation.
    - Surgical Methods: Vasectomy in males and tubectomy in females block reproductive pathways, offering reliable prevention.
- **Abortion and Female Foeticide:** While surgical abortion is used to terminate unwanted pregnancies, it can be misused for illegal female foeticide. Laws prohibit prenatal sex determination to maintain a balanced sex ratio and promote a healthy population.